BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MIRPURKHAS

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQ's)

Note: Q.1 Choose the correct answer for each from the given options.

Qi. $(A^c)^c$ = _____

(a) (A')' (b) A'

(c) A (d) B

Qii. Which operation is not commutative

(a) Symmetric difference (b) Union

(c) Difference (d) Intersection

Qiii. A function is called into if range co-domain

(a) Q (b) -

(c) = (d) none of these

Qiv. A function is called surjection if range _____ co-domain.

(a) $\supseteq$ (b) $\subset$

(c) = (d) alll of them

Qv. Which operation is not commutative

(a) Symmetric difference (b) Union

(c) Difference (d) Intersection

Qvi. If (x, 6) = (2, y - 6) then x+ y -

(a) 8 (b) 10

(c) 12 (d) 14

Qvii. If a: b = c : d then according to alternando property:

(a) a/b = c/d
(b) (a+b)/b = (c+d)/d
(c) [illegible in source scan]
(d) a/c = b/d

Qviii. If O(A×B) = 100 and O(B) = 5 then O(A) -

(a) 10
(b) 15
(c) 20
(d) 25

Qix. A function is called surjective if range _____ co-domain

(a) Q
(b) -
(c) =
(d) all of these

Qx. If A = 1, 2, 3, 41 and B ={2,4,6} then AΔB _____

(a) {1,2}
(b) {6}
(c) {1,3}
(d) {1,3,6}

Qxi. The conclusion of the enunciation "If two chords of a circle are congruent, then they are equidistant from the centre." is:

(a) two chords of the circle are equidistant from the centre.
(b) two chords of a circle are congruent.
(c) a circle has two chords.
(d) the centre of circle is equidistant from chords.

Qxii. In the adjacent figure, the point of tangency is .

(a) A
(b) B
(c) C
(d) D

Qxiii. Tangents drawn at end points of a diameter of a circle are __ to each other

(a) parallel
(b) perpendicular
(c) intersecting
(d) both (b) and (c)

Qxiv. The maximum number of common tangents between two circles touching internally is

(a) 0
(b) 1
(c) 2
(d) 3

Qxv. The maximum number of common tangents between two circles touching externally is

(a) 0
(b) 1
(c) 2
(d) 3

Qxvi. If the sum of the roots of $(p +1)x^2+ (2p + 3) x + (3 p +4) = 0$ is -1, then product of the root is

(a) 0
(b) 1
(c) 2
(d) 3

Qxvii. $(x^2 + 1) / ((x - 1)(x + 2))$ is,

(a) Proper fraction

(b) An improper fraction

(c) An identity

(d) A constant term

Section-B

> Note: Attempt any 08 question of the following:

Q2. If U = {1,2,3,...10}, A = {1,2,3,...,5} and B = {1,3,5,7,9} then find:

(a) A'

(b) B'

(c) A∩B'

(d) A∪B

Q3. Verify De Morgan's laws using venn diagram if A={1,3,5,7,9}, B={5,6,7,8}, and U={1,2,3,.....10}

Q4. Find the values of x and y if:

(i) (x - 5,10) = (11,y - 7)

(ii) (5x+8,5y - 4) = (3x+10,2y +2)

(iii) (2x -3y,5x+y) = (3,16)

Q5. The cube root of x varies as the square of y if x = 8 when y = 3 find x when y = 3/2.

Q6. prove that: a:b = c:d if

(a) $a^2 - b^2 : a^2 + b^2 = ac - bd + ac + bd$

(b) (a+b+c+d)/(a+b-c-d) = (a-b+c-d)/(a-b-c+d)

Q7. What is the order of each of the following matrix?

(i) [0 1 -1; 2 0 3; -1 -3 2]

(ii) [3 4; -1 2; 0 3]

(iii) [1 1 1; 2 2 2]

(iv) [-2; 1; -1]

(v) [0]

Q8. Show that the roots of the following quadratic equation are real: $x^2 - 2x(k + 1/k)x + 3 = 0$, $\forall k \in \mathbb{R} - \{0\}$

Q9. $(x-1)^2 + (y + 3)^2 = 25$ and $x + (y + 1)^2 = 10$

Q7. Resolved: $(3x^3 - 2x^2 - 16x + 20) / ((x - 2)(x + 2))$

Q8. Find average attitude for dogs using A.M medain and mode from 60 people data.

Attitude: Love dogs / Like dogs / No Option / Dislike dogs / Hate dog — No: of people: 20 / 15 / 4 / 13 / 8

Q9. Form the equation whose roots are:

(i) 2, -3

(ii) 2+i, 2-i

Q10. If α,β are the roots of the equation $px + qx + q = 0$, find the value of $\sqrt{(\alpha/\beta)} + \sqrt{(\beta/\alpha)}$

Q11. The foot of a ladder is placed 6 feet from a wall. if the top of the ladder rest 8 feet up on the wall. How long is the ladder?

Q12. Find the cube roots of 729.

Section-C

> Note: Attempt any 04 Question of the following:

Q13. In the folloiwng $m\angle PQR = 35°$ and $m\angle PRQ = 31°$. Find $m\angle QPR$ and $m\angle OQR$.

Q14. Specify the type of each of the following matrixs.

(i) [2 -1; 3 0]

(ii) [3; 4]

(iii) [-2 0]

(iv) [λ 0; 0 λ]

(v) [a 0 0; 0 b 0; 0 0 c]

Q15.

(a) Show that the diameters of a circle bisect each other.

(b) Find distance between two parallel chords of length α and β in a circle of radius r for the following give values of α, β and r.

(a) $\alpha = 6cm$, $\beta = 8cm$, $r = 5cm$ (b) $\alpha = 3cm$, $\beta = 6cm$, $r = 14cm$

Q16. Shwo that the rectangle circumscribeing a circle must be a square.

Q17. if length of the segment joining centres, of two circles in the sum of their ratii, show that the circles touch esternally.

Q18. Find the values(s) of k that ensure that following quadratic equation have:

(a) Same solution

(b) Different and solutions